

Lab 12: Fast LRC Decay

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Pre-lab Questions

Derive the relationship between the time constant in an LRC circuit and the half life of the voltage across the resistor.

From the general solution for the voltage across the resistor in an LRC circuit, we have:

$$V(t) = V_0 e^{-\frac{R}{2L}t} \cos(\omega t + \varphi),$$

where $\omega = \sqrt{\frac{1}{LC} - \frac{R^2}{4L^2}}$ is the angular frequency of the oscillation. The time constant τ is defined as the time it takes for the voltage to decay to $1/e$ of its initial value, which occurs when the exponential is $1/e$:

$$e^{-\frac{R}{2L}\tau} = \frac{1}{e} \rightarrow -\frac{R}{2L}\tau = -1 \rightarrow \tau = \frac{2L}{R}.$$

This allows us to rewrite the voltage as:

$$V(t) = V_0 e^{-t/\tau} \cos(\omega t + \varphi).$$

The half-life $t_{1/2}$ is defined as the time it takes for the voltage to decay to half of its initial value, which occurs when the exponential is $1/2$:

$$e^{-t_{1/2}/\tau} = \frac{1}{2} \rightarrow -t_{1/2}/\tau = \ln\left(\frac{1}{2}\right) \rightarrow t_{1/2} = \tau \ln 2 = \frac{2L \ln 2}{R}.$$